

```
~ subfinder -d hackerone.com -o subdomains.txt

projectdiscovery.io

[INF] Current subfinder version v2.6.7 (outdated)
[INF] Loading provider config from /home/an0nbil/.config/subfinder/provider-config.yaml
[INF] Enumerating subdomains for hackerone.com
ebiz.hackerone.com
gslinkq.hackerone.com
widget.hackerone.com
auth.hackerone.com
iwu.hackerone.com
work.hackerone.com
win.hackerone.com
```

Subdomain Discovery_ Passive+Active Sub Discovery

Name: FIREWALL FAIRY

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1. Overview

Subdomain Discovery is the process of finding subdomains of a target domain. During this practical, I used two tools:

- **Subfinder** – Finds subdomains from passive online sources.
- **PureDNS** – Checks which discovered subdomains are actually valid and resolve through DNS.

Workflow

```
Target Domain
↓
Subfinder
```

```
↓  
sub.txt (All Found Subdomains)  
↓  
PureDNS  
↓  
valid.txt (Only Valid Subdomains)
```

2. Objective

- Find subdomains of a target domain.
- Save all discovered subdomains.
- Verify which subdomains are valid.
- Save only valid subdomains for the next reconnaissance step.

3. Installation

Install Subfinder

```
sudo apt install subfinder
```



```
(kali㉿kali)-[~]  
└─$ sudo apt install subfinder  
Installing:  
subfinder
```

Verify installation

```
subfinder-h
```

```
(kali㉿kali)-[~]  
$ subfinder -h  
Subfinder is a subdomain discovery tool that discovers subdomains for websites by using passive online sources.
```

```
(kali㉿kali)-[~]  
$ subfinder -ls  
remote: Enumerating objects: 1231, done.  
remote: Counting objects: 100% (715/715), done.  
remote: Compressing objects: 100% (516/516), done.  
remote: Total 1231 (delta 529), reused 496 (delta 496), pack-reused 516 (from 1)  
projectdiscovery.io
```

4. Install PureDNS

First, install Go.

```
sudo apt updatesudo apt install golang-go
```

```
(kali㉿kali)-[~]  
$ sudo apt update  
sudo apt install golang-go  
[sudo] password for kali:
```

Install PureDNS.

```
go install github.com/d3mondev/puredns/v2@latest
```

```
(kali@kali)-[~]  
$ go install github.com/d3mondev/puredns/v2@latest
```

Install MassDNS (required by PureDNS).

```
git clone https://github.com/blechschmidt/massdns.git  
cd massdns  
make install
```

```
(kali@kali)-[~]  
$ git clone https://github.com/blechschmidt/massdns.git  
cd massdns  
make  
sudo make install
```

5. Issue I Faced

After installing PureDNS, I tried:

```
puredns --version
```

Error:

```
puredns: command not found
```

```
(kali㉿kali)-[~]  
$ puredns resolve sub.txt -r ~/.config/puredns/resolvers.txt  
puredns: command not found
```

Reason

The Go binary folder (`~/go/bin`) was not added to the system PATH.

Solution

Check if PureDNS was installed.

```
ls ~/go/bin
```

Run it directly.

```
~/go/bin/puredns --version
```

Output

```
puredns version v2.1.1
```

Add Go binary to PATH.

```
echo 'export PATH="$HOME/go/bin:$PATH"' >> ~/.zshrc
```

```
source ~/.zshrc
```

Verify.

```
which puredns
```

Result

```
/home/kali/go/bin/puredns
```

Now the command works.

```
puredns --version
```

Output

```
puredns version v2.1.1
```



```
(kali㉿kali)-[~]  
$ puredns --version  
puredns version v2.1.1
```

6. Practical Test

Step 1 – Find Subdomains

Target:

```
hackerone.com
```

Command:

```
subfinder-d hackerone.com
```

Result:

Example:

```
(kali㉿kali)-[~]
$ subfinder -d hackerone.com

      _ _ _ _ _
     / / / / /
    / / / / /
   / / / / /
  / / / / /
 / / / / /
/ / / / /

projectdiscovery.io

[INF] Current subfinder version v2.14.0 (latest)
[INF] Loading provider config from /home/kali/.config/subfinder/provider-config.yaml
[INF] Enumerating subdomains for hackerone.com
ns2.hackerone.com
docs.hackerone.com
a.ns.hackerone.com
www.hackerone.com
mail2.hackerone.com
managed.hackerone.com
```

Command

Purpose

Save all discovered subdomains into a file.

Check the file.

```
cat sub.txt
```

Result

sub.txt contains **24 discovered subdomains**.

Step 3 – Download Resolver List

PureDNS requires DNS resolvers.

Create the configuration folder.

```
mkdir -p ~/.config/puredns
```

Download the resolver list.

```
wget https://raw.githubusercontent.com/trickest/resolvers/main/resolvers.txt \
-O ~/.config/puredns/resolvers.txt
```

Check the file.

```
head ~/.config/puredns/resolvers.txt
```

```
(kali㉿kali)-[~]
$ head ~/.config/puredns/resolvers.txt
99.99.99.193
99.92.207.51
99.83.147.72
99.79.50.252
99.79.182.170
99.42.199.193
98.26.23.128
98.213.193.14
98.213.192.240
98.213.192.223
```

Step 4 – Validate Subdomains

Command

```
puredns resolve sub.txt \
-r ~/.config/puredns/resolvers.txt
```

PureDNS checked all discovered subdomains and removed invalid ones.

Result

- **24** subdomains received from Subfinder.
- **10** valid subdomains found after DNS validation.

Example

```
www.hackerone.com
api.hackerone.com
docs.hackerone.com
support.hackerone.com
```

```
Found 10 valid domains:
```

```
b.ns.hackerone.com
```

```
www.hackerone.com
```

```
docs.hackerone.com
```

```
api.hackerone.com
```

```
a.ns.hackerone.com
```

```
mta-sts.hackerone.com
```

```
mta-sts.managed.hackerone.com
```

```
gslink.hackerone.com
```

```
support.hackerone.com
```

```
mta-sts.forwarding.hackerone.com
```

Step 5 – Save Valid Subdomains

Command

```
puredns resolve sub.txt \  
-r ~/.config/puredns/resolvers.txt \  
-w valid.txt
```

Check the output.

```
cat valid.txt
```

Result

The file contains only valid subdomains.


```
puredns --version

# Validate subdomains
puredns resolve sub.txt \
-r ~/.config/puredns/resolvers.txt

# Save valid subdomains
puredns resolve sub.txt \
-r ~/.config/puredns/resolvers.txt \
-w valid.txt

# View results
cat sub.txt
cat valid.txt
```